Structures and Unions Assessment

Name: ______________________________

Registration Number: ______________________________

Institute: ☐ KCE ☐ KIT ☐ KAHE

1.
```
struct Student{
    int id;
    char grade;
};
```
What does `sizeof(struct Student)` typically return on a 32-bit system?

- ○ 5
- ○ 6
- ○ 8
- ○ 9

2. Which operator is used to access a member through a pointer?

- ○ .
- ○ ::
- ○ ->
- ○ *

3. `struct Student s; struct Student *p=&s;` **Which statement is correct?**

- ○ p.id
- ○ p->id
- ○ (*p)->id
- ○ s->id

4. Which operator is equivalent to `ptr->age`?

- *ptr.age
- (*ptr).age
- ptr.age
- &(ptr.age)

5. `struct A{ char c; int i; };` Padding is added because

- C language requirement
- Compiler optimization for alignment
- To reduce RAM
- To increase variables

6. Which occupies less memory?

- Structure
- Union
- Depends on compiler
- Both same

7. A union stores

- Every member simultaneously
- Only one member at a time
- Two members together
- No members

8. The size of a union is

- Sum of member sizes
- Largest member size (plus alignment)
- Smallest member size
- Zero

9. `union Data{ int i; float f; };` Writing to `i` and reading `f`

- ○ Always correct
- ○ Undefined/implementation-dependent behavior
- ○ Compile error
- ○ Runtime error

10. Which declaration creates a pointer to structure?

- ○ struct S *p;
- ○ struct *S p;
- ○ pointer struct S;
- ○ S->p;

11. `struct Node{ struct Node *next; };` This is called

- ○ Nested structure
- ○ Self-referential structure
- ○ Anonymous structure
- ○ Packed structure

12. `struct Node{ struct Node next; };` This declaration is

- ○ Correct
- ○ Incorrect
- ○ Warning only
- ○ Compiler dependent

13. Structures are passed to functions

- ○ By reference
- ○ By value
- ○ By pointer automatically
- ○ None

14. Passing a large structure to a function is

- ○ Faster
- ○ Slower due to copying
- ○ Impossible
- ○ Same as pointer

15. Which is more efficient?

- ○ Pass structure
- ○ Pass pointer to structure
- ○ Both same
- ○ None

16. `struct Student *p; printf("%lu",sizeof(p));` Output on a 64-bit system

- ○ 4
- ○ 8
- ○ Size of structure
- ○ Depends on members

17. `printf("%lu",sizeof(*p));` returns

- ○ Pointer size
- ○ Structure size
- ○ Integer size
- ○ Compile error

18. Structure assignment `s2=s1;`

- ○ Illegal
- ○ Legal
- ○ Runtime error
- ○ Only for integers

19. Can two structures be compared using `==`?

- Yes
- No
- Only integers
- Only floats

20. Which is true?

- Structures support assignment
- Structures support comparison
- Structures support multiplication
- Structures support increment

21. Which memory is allocated by `malloc(sizeof(struct Student))`

- Stack
- Heap
- Register
- ROM

22. `struct Student *p=NULL; p->id=5;` Result

- Works
- Segmentation fault/Undefined behavior
- Compile error
- Prints 5

23. `struct Student s[10];` This declares

- Pointer
- Array of structures
- Structure pointer
- Union

24. Access 5th student's age

- ○ s[4].age
- ○ s.age[4]
- ○ age.s[4]
- ○ s->age

25. `struct Student *p=s; p+1` points to

- ○ Next byte
- ○ Next structure
- ○ Next integer
- ○ NULL

26. Pointer arithmetic on structure pointer increments by

- ○ 1 byte
- ○ Size of structure
- ○ Size of int
- ○ Size of pointer

27. A structure can contain

- ○ Another structure
- ○ Array
- ○ Pointer
- ○ All of these

28. Which keyword creates an alias?

- ○ define
- ○ typedef
- ○ alias
- ○ using

29. `typedef struct{ int x; }Point;` **Variable declaration**

- Point p;
- struct Point p;
- typedef p;
- Point->p

30. Unions are mainly used for

- Increasing memory
- Saving memory
- Dynamic memory
- Virtual memory

31. Embedded systems commonly use unions for

- Memory sharing
- File handling
- Graphics
- Sorting

32. Hardware registers are often represented using

- Structures
- Strings
- Arrays only
- Floats

33. Bit-fields are commonly combined with

- Arrays
- Unions
- Functions
- Macros only

34. Which statement copies all members?

- memcpy()
- Assignment operator
- Both A and B
- None

35. Returning pointer to local structure `return &s;` is

- Correct
- Dangerous
- Faster
- Required

36. Structure members are stored

- Randomly
- Sequentially (with possible padding)
- Reverse order
- Alphabetically

37. Which statement accesses second structure in array? `p->id`

- p[1].id
- p.id[1]
- (*p)[1]
- p->id[1]

38. Which keyword removes padding (compiler-specific)?

- packed
- align
- pragma
- volatile

39. Structure alignment improves

- ○ Memory usage
- ○ CPU performance
- ○ Source code size
- ○ File size

40. Linked lists are implemented using

- ○ Arrays
- ○ Self-referential structures
- ○ Unions
- ○ Strings

41. Trees are implemented using

- ○ Self-referential structures
- ○ Arrays only
- ○ Unions only
- ○ Integers

42. `char name[20]; };` **name is**

- ○ Pointer
- ○ Array
- ○ Structure
- ○ Union

43. Which function allocates contiguous memory?

- ○ malloc()
- ○ printf()
- ○ scanf()
- ○ strcpy()

44. `free(p); p->id=10;` Result

- ○ Valid
- ○ Undefined behavior
- ○ Compile error
- ○ Works on Linux only

45. The arrow operator (`->`) combines

- ○ * and .
- ○ & and .
- ○ + and .
- ○ * and &

46. Which member determines the minimum size of a union?

- ○ First member
- ○ Last member
- ○ Largest member
- ○ Smallest member

47. Can a structure contain function pointers?

- ○ Yes
- ○ No
- ○ Only C++
- ○ Only Linux

48. Linux device drivers often use structures containing

- ○ Function pointers
- ○ Strings
- ○ Arrays only
- ○ Floats

49. Which copy duplicates only pointer addresses?

- ○ Deep copy
- ○ Shallow copy
- ○ Smart copy
- ○ Recursive copy

50. For a UART packet containing multiple fields and different interpretations of the same bytes, the best design is

- ○ Only arrays
- ○ Only pointers
- ○ Structure + Union
- ○ Only integers